Word Order in News Texts: Pre vs Post ChatGPT

Management Summary

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January 24, 2026

1. Overview

The goal of this project was to investigate how sentence word order can be automatically identified across multiple languages using computational methods. Specifically, we focused on identifying the relative order of **subject, verb, and object** (e.g., Subject–Verb–Object) in sentences from **English, German**, and **Russian** news texts.

To address this task, we developed a multilingual processing pipeline that combines linguistic analysis and machine learning. We implemented both rule-based and machine learning models and evaluated their performance against stanza-annotated subject-verb-object order (ground truths extracted from stanza). Our results show moderate overall accuracy with strong variation between languages. Crucially, the project revealed that most errors stem from limitations in the extraction of grammatical roles rather than from the classification models themselves.

2. Task Description and Challenges

2.1 Task Description

The task was to automatically classify the word order of sentences into classes such as Subject–Verb–Object (SVO), Subject–Object–Verb (SOV), and related variants. Each sentence had to be analyzed to determine which words function as the subject, verb, and object, and then classify their relative order.

For each language, five years were selected with the aim of covering proportional periods before and after the introduction of ChatGPT in November 2022, subject to data availability. This selection enables a comparison of word order patterns across time and between the pre- and post-ChatGPT periods.

2.2 Key Challenges

This task presents several challenges, especially in a multilingual setting:

- **Cross-linguistic variation:** English, German, and Russian differ significantly in the degree to which word order can vary freely.
- **Complex sentence structures:** News texts often contain multiple subordinate clauses, passive constructions, and auxiliary verbs.
- **Ambiguity in grammatical roles:** Automatically identifying the "main verb" or the object is not always straightforward.
- **Incomplete or partial orders**: The majority of the sentences were without verbs or objects.

These challenges necessitated careful evaluation not only of final predictions but also of the quality of intermediate linguistic annotations.

3. Data and External Resources

The project used multilingual news-sentence datasets from 5 years for English, Russian, and German (10,000 sentences/year) from the Leipzig Corpora Collection after pre-processing, split into training, validation, and test sets. For evaluation purposes, a subset of **60 sentences** (20 per language) from the test set was manually annotated with gold-standard word order labels.

We relied on the following external resources:

- **Stanza**, an open-source linguistic analysis toolkit, for automatic syntactic annotation
- Standard Python-based machine learning libraries for model training and evaluation

All tools used are publicly available and widely adopted in academic research.

4. Solution Overview

4.1 Linguistic Annotation

First, each sentence was automatically analyzed using the Stanza model for the respective languages to identify its parts of speech components (subject, verb, object, etc.). This step provided the structural information required for word order classification. We extracted the word order from these annotations using a rule.

4.2 Rule-Based Baselines

We implemented two simple rule-based classifiers as a baseline. This approach relied on basic positional heuristics, such as identifying whether nouns appear before or after the first verb. While this method is linguistically naive, it provides a transparent reference point and highlights the difficulty of the task.

4.3 Machine Learning Classifiers

The analysis evaluated three supervised machine learning models: Support Vector Machines (SVM), Naive Bayes, and Logistic Regression. For each model, two feature extraction methods were applied: Bag-of-Words and TF–IDF representations of the text.

All models were trained using hyperparameter tuning and compared using the same evaluation procedure. The results showed that the overall performance across models was comparable. Logistic Regression combined with TF–IDF features achieved marginally better results and was therefore selected as the final model.

5. Evaluation

To assess Stanza's annotation quality and the downstream predictions of the Rule-based and ML classifiers, we conducted a manual evaluation using 20 sentences per language. A prediction was considered correct only if the extracted word order exactly matched the manually annotated gold standard.

5.1 Manual Annotation - Results

The overall accuracy of the stanza annotations is 55%(33/60 correct). Russian annotations are the most reliable, with an accuracy of 70%, and English annotations are 65% accurate, while German annotations have only 30% accuracy. These results indicate big language-specific differences.

6. Error Analysis and Limitations

The evaluation revealed that most errors originate from the **extraction of parts of speech**, not from the classification logic itself:

- **Russian:** Stanza performs reliable in comparison with other observed languages, resulting in relatively accurate word order extraction.
- **English:** Errors mainly involve failure to extract objects correctly, especially in complex noun phrases.
- **German:** The primary issue is the incorrect identification of auxiliary verbs as main verbs. This leads to systematic misclassifications of word order in constructions involving passive voice, perfect tense, and modal verbs.

These findings show that the main limitation lies in the preprocessing stage rather than in the predictive models.

7. Insights

7.1 Word Order Distribution

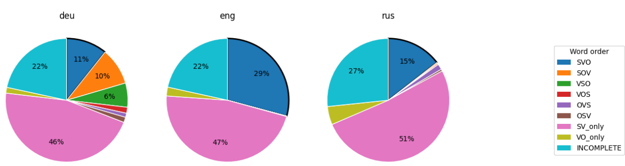
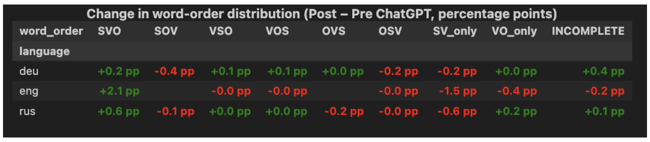


Figure 1: Word order in the observed languages across the full time period

English, known for its fixed order, shows the highest prevalence of SVO sentence order (29%) compared to the other languages. German is known for its more flexible word order, governed by the verb-second (V2) rule. This rule is partially reflected in the data, as SV_only, INCOMPLETE, and SVO are the most frequent classes. However, the presence of SOV (10%), VSO (8%), VOS (1%), and OSV (1%) structures represents deviations from this constraint. Russian is expected to exhibit a flexible word order, and the observed distribution does not contradict this assumption.

7.2 Change in Word Order



Change in word-order distribution (Post – Pre ChatGPT, percentage points)

word_order language	SVO	SOV	VSO	VOS	OVS	OSV	SV_only	VO_only	INCOMPLETE
deu	+0.2 pp	-0.4 pp	+0.1 pp	+0.1 pp	+0.0 pp	-0.2 pp	-0.2 pp	+0.0 pp	+0.4 pp
eng	+2.1 pp		-0.0 pp	-0.0 pp		-0.0 pp	-1.5 pp	-0.4 pp	-0.2 pp
rus	+0.6 pp	-0.1 pp	+0.0 pp	+0.0 pp	-0.2 pp	-0.0 pp	-0.6 pp	+0.2 pp	+0.1 pp

Figure 2: Changes in the prevalence of word order types following the introduction of ChatGPT in the observed languages

All languages show a slight increase in the prevalence of SVO word order following the introduction of ChatGPT, with English showing the most prominent difference. In terms of stability, English shows changes primarily in the SVO class and the related partial classes SV_only, VO_only, and INCOMPLETE. In contrast, German and Russian also exhibit minor changes in other word order classes (SOV, VSO, VOS, and OVS). However, the overall change is higher in English compared to the other two languages.

8. Next Steps

Future work should focus on addressing the identified methodological limitations to improve the accuracy of sentence labeling, particularly with respect to main clause identification and partial word orders. Conducting significance tests on the observed increase in SVO word order in Russian and German would help determine whether the effect is statistically meaningful or due to chance. Expanding the dataset would allow for a more comprehensive analysis and more robust conclusions.

9. Contributions

Assylbek Tleules: Pre-processing, Repository maintenance, Gold Standard Stanza annotation, Rule-based baselines, Russian language manual annotations.
Emily Jacob: Error Analysis, Manual annotation, Pre-processing & Presentation
Luka Santek: Machine learning models, Analysis of word order distributions (Insights & Visualizations)
Tehseen Ali Tahir: Research, Presentation, Datasets management